9.6 Permutations and Combinations

Name _____ Period ____ Date _____

Algebra I 9.6 (30-minute plan) · Student sheet

I can…

- count the ways to arrange objects using factorials.
- find permutations and combinations and tell which one to use.
- use permutations and combinations to find probabilities.

Vocabulary · fill in the missing word

n factorial (n!) — $n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$; 0! = _____

Permutation — an _____ in which order matters

$_nP_r$ — $n!/(n-r)!$, the number of _____ of r objects chosen from n

Combination — a _____ in which order does not matter

$_nC_r$ — $n!/((n-r)! \cdot r!)$, the number of _____ of r objects chosen from n

Part 1 · Notes and examples (fill in each note, then work the example)

Factorial: n! = _____ 0! = _____

Example 1

a. In how many ways can you arrange the letters in MATH?

b. In how many ways can you arrange 2 of the letters?

Permutations: Order matters → _____: $_nP_r$ = _____

Example 2

Ten runners are in a race. In how many ways can the gold, silver, and bronze medals be awarded?

Combinations: Order doesn't matter → _____: $_nC_r$ = _____

Probability with counting: Probability = (favorable selections) ÷ _____

Example 3

A pizza shop has 8 toppings. How many different 3-topping pizzas can you order?

Part 2 · Partner practice

Partner practice

a. In how many ways can 6 books be arranged on a shelf?

b. In how many ways can 3 of the 6 books be arranged?